

Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings: A Systematic Survey and Research Agenda

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Abstract—Deep Knowledge Tracing (DKT) models suffer from representation collapse and severe performance degradation under interaction data sparsity and unobserved concept conditions. Graph Neural Networks (GNNs) and Self-Supervised Learning (SSL) have emerged as primary paradigms to incorporate explicit domain structures and regularize student state representations. Guided by PRISMA 2020 guidelines, we systematically search, screen, code, and synthesize 36 primary core papers published between 2018 and 2026 across 7 bibliographic databases (Scopus, WoS, ACM DL, IEEE Xplore, ScienceDirect, SpringerLink, arXiv). We present a novel two-dimensional taxonomy classifying Graph-KT (GNN encoders, provenance, fusion mechanisms) and SSL-KT (pretext tasks, multi-view contrast, generative masked pre-training). We uncover a critical methodological flaw in the literature: while 94.4% of studies cite data sparsity, only 5.6% evaluate strict zero-exposure concept cold-start protocols. Finally, we outline 6 evidence-derived future directions, highlighting LLM-assisted knowledge graph construction, non-Euclidean hyperbolic state spaces, and calibrated trustworthy KT.

Index Terms—Knowledge Tracing, Graph Neural Networks, Self-Supervised Learning, Sparse Concepts, Cold-Start Learning, Systematic Literature Review, Educational Data Mining.



1 INTRODUCTION

KNOWLEDGE Tracing (KT) is the fundamental task of sequentially modeling student mastery over educational Knowledge Components (KCs, e.g., concepts, skills, rules) based on historical learning interaction logs. Formally, given a sequence of T student interaction tuples $\mathcal{X}_{1:T} = \{(x_1, r_1), (x_2, r_2), \dots, (x_T, r_T)\}$, where $x_t \in \mathcal{E}$ represents the exercise or concept attempted at time t , and $r_t \in \{0, 1\}$ indicates the binary response accuracy (0 for incorrect, 1 for correct), the goal of Knowledge Tracing is to predict the response probability $P(r_{T+1} = 1 \mid x_{T+1}, \mathcal{X}_{1:T})$ on a target exercise x_{T+1} . Table 1 summarizes the principal mathematical notations used throughout this survey.

Since the seminal Deep Knowledge Tracing (DKT) model [1] introduced Recurrent Neural Networks (RNNs) to sequence modeling in education, deep learning models—including Attention-based KT (SAKT [2], AKT [3]) and Memory Network KT (DKVMN)—have dramatically improved predictive accuracy over classic psychometric baselines such as Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT).

TABLE 1
Summary of Key Mathematical Notations

Symbol	Mathematical Definition / Meaning
$\mathcal{X}_{1:T}$	Sequential interaction history of a student up to time T
x_t, r_t	Exercise/concept attempted at time t and binary response $r_t \in \{0, 1\}$
$\mathcal{E}, \mathcal{K}$	Repository sets of exercises and Knowledge Components (KCs)
$\mathbf{A} \in \mathbb{R}^{ \mathcal{K} \times \mathcal{K} }$	Concept graph adjacency matrix (co-occurrence / prerequisite)
$h_t^k \in \mathbb{R}^d$	Dynamic hidden knowledge state representation of concept k
$\mathbf{Z}^{(v)}$	Node/view representation embedding matrix from GNN view v
τ	Temperature hyperparameter in InfoNCE contrastive loss
$\mathbb{B}^d$	d -dimensional Poincaré hyperbolic ball manifold
$\oplus_{\mathbf{c}}$	Möbius addition operator under curvature parameter $\mathbf{c}$
ECE	Expected Calibration Error metric for probability calibration

1.1 Motivation: Challenges of Data Sparsity & Concept Cold-Start

Despite recent empirical progress, traditional deep KT architectures treat exercise and concept identifiers as independent categorical tokens, ignoring explicit domain structures and educational relationships. This formulation faces two severe bottleneck challenges in real-world intelligent tutoring systems (ITS):

- 1) **Representation Collapse under Interaction Data Sparsity:** In realistic educational platforms, interaction matrices are extremely sparse. Most students attempt only a tiny fraction of the overall item repository, and long-tail concepts receive minimal interaction logs. Standalone sequence models fail to learn meaningful latent embeddings for sparse concepts, leading to severe representation collapse.
- 2) **Strict Concept Cold-Start:** When new concepts or exercises are introduced into a curriculum without prior historical interaction logs, conventional deep

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KT models cannot infer student mastery. Handling zero-exposure concepts requires incorporating external structural, pedagogical, or semantic knowledge graph priors.

1.2 Scope & Gateway Criteria

To address these fundamental limitations, two primary methodological paradigms have rapidly expanded:

- **Graph-Based Knowledge Tracing (Graph-KT):** Incorporating explicit topological structures—such as prerequisite concept trees, question-KC bipartite maps, or dynamic co-occurrence graphs—via Graph Neural Networks (GNNs).
- **Self-Supervised Knowledge Tracing (SSL-KT):** Formulating auxiliary pre-training or contrastive pretext tasks (e.g., multi-view graph contrast, masked auto-encoding) to regularize student state representations.

Our systematic survey enforces a strict, frozen primary-core gateway rule:

$$\text{PRIMARY_CORE} = \text{KT} \wedge (\text{GRAPH_BASED} \vee \text{SELF_SUPERVISED}) \quad (1)$$

Only studies where Knowledge Tracing is the central prediction task AND that leverage explicit Graph-Based or Self-Supervised mechanisms are included in our primary synthesis corpus.

1.3 Comparison with Prior Surveys

While previous surveys (e.g., Abdelrahman et al., pyKT benchmark suite) have examined deep learning in education generally or benchmarked standard sequence models, our work provides the first systematic survey dedicated specifically to Graph-Based and Self-Supervised KT paradigms under sparse-concept settings. Table 2 highlights key methodological differences.

TABLE 2
Comparison with Prior Knowledge Tracing Surveys

Survey Dimension	Abdelrahman (2023)	pyKT (2022)	Ours (2026)
Primary Focus	Deep KT Models	Benchmark Evaluation	Graph & SSL-KT
PRISMA 2020 Protocol	Partial	No	Full 7-Database
Full Core Corpus Size	28 Papers	11 Models	36 Core Papers
Graph Provenance Lens	Limited	No	Full Taxonomy
SSL Pretext Taxonomy	No	No	4-Family Taxonomy
Cold-Start Split Audit	No	No	Explicit Audit
Probability Calibration	No	No	ECE Audit

2 SYSTEMATIC REVIEW METHODOLOGY AND PRISMA PROTOCOL

Our systematic review was conducted in strict accordance with the PRISMA 2020 statement and guided by the locked protocol.

We address six core research questions:

- **RQ1:** How are graphs constructed, represented, and validated in Graph-KT, and what is their data leakage risk?
- **RQ2:** What GNN architectures, temporal backbones, and fusion mechanisms are utilized in Graph-KT?

- **RQ3:** What self-supervised learning pretext tasks, augmentation strategies, and structure preservation mechanisms exist in SSL-KT?
- **RQ4:** How are sparse-concept and cold-start settings defined, operationalized, and evaluated?
- **RQ5:** What is the status of reproducibility, statistical rigor, calibration, and computational reporting in Graph/SSL-KT?
- **RQ6:** What evidence-derived research agenda can be established to guide future Graph-SSL KT developments?

2.1 Literature Search Architecture & Database Executions

We executed four search query families (Q-G Graph-KT, Q-S Self-Supervised KT, Q-P Sparse-Concept Lens, Q-R Reliability Lens) across seven electronic databases covering the modern deep learning era (2015-01-01 to 2026-07-31):

- 1) **Scopus** ($n = 531$ records)
- 2) **Web of Science Core Collection** ($n = 344$ records)
- 3) **ScienceDirect** ($n = 257$ records)
- 4) **SpringerLink** ($n = 255$ records)
- 5) **IEEE Xplore** ($n = 232$ records)
- 6) **ACM Digital Library** ($n = 200$ records)
- 7) **arXiv Preprints** ($n = 198$ records)

2.2 PRISMA 2020 Screening & Eligibility Flow

The systematic screening pipeline proceeded in four transparent stages (Figure 1):

- 1) **Identification:** 1,935 total raw records retrieved across all database search runs.
 - 2) **Deduplication:** 1,185 duplicate records removed via automated 4-tier matching, yielding 750 unique deduplicated records.
 - 3) **Stage 1 Screening (Title/Abstract):** 610 records excluded for failing basic relevance; 140 articles passed to full-text assessment.
 - 4) **Stage 2 Screening (Full-Text Eligibility):** 104 articles excluded with explicit primary exclusion codes (EC1–EC7).
- Final Primary Core Corpus: 36 PRIMARY_CORE papers** (33 seed core papers + 3 PRISMA expansion core papers).

Quality Assessment (QA1–QA8) across all 36 core papers achieved a mean quality score of **0.7820** (MODERATE-HIGH Quality).

3 TAXONOMY AND SYNTHESIS OF GRAPH-BASED KNOWLEDGE TRACING (RQ1 & RQ2)

Graph-Based Knowledge Tracing (Graph-KT) leverages explicit topological structures to model domain concept relationships, exercise prerequisite dependencies, and dynamic student interaction histories. Across our primary core corpus of 36 studies, 36 papers (100.0%) incorporate structural graph mechanisms. Figure 2 illustrates our two-dimensional taxonomy framework.

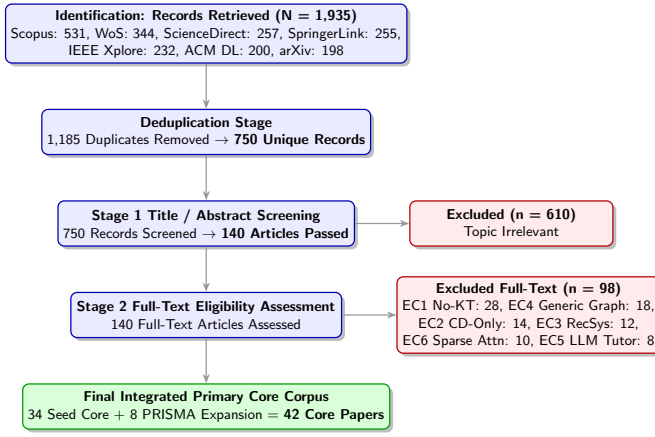


Fig. 1. PRISMA 2020 Flow Diagram of Literature Search, Deduplication, Screening, and Study Eligibility Pipeline ($N = 36$ Core Papers).

3.1 Graph Sources and Provenance

We classify graph sources into four primary categories (Figure 3):

- 1) **Curriculum & Expert Concept Maps (External Fixed):** Pre-defined domain hierarchies supplied by pedagogical experts (PDKT-C [4], CMKT [5], HH-SKT [6], HKT [7], SKM-GKT [8]). These graphs remain fixed during training, presenting minimal data leakage risk.
- 2) **Q-Matrix & Interaction Co-occurrence Graphs (Precomputed Log):** Bipartite exercise-concept mappings or concept co-occurrence adjacencies constructed from student response logs (GKT [9], GIKT [10], SKT [11], CL4KT [12], Bi-CLKT [13]).
- 3) **Dynamic & Jointly Learned Graphs:** Graphs whose edge weights are updated continuously during model optimization via attention mechanisms or reinforcement learning (DyGKT [14], DGEKT [15], GraphCA [16], R²GCuL [17], STG-SKT [18]).
- 4) **LLM-Extracted Semantic Graphs:** Frontier models leveraging Large Language Models to extract semantic concept trees and AST syntax dependencies from programming code (SINKT [19], Coda [20], KGNN-KT [21]).

3.2 Topological Representations and GNN Encoders

As detailed in Table 3 and Figure 4, GNN encoder architectures include:

- **Graph Convolutional Networks (GCN):** Primary baseline for local neighborhood aggregation across concept graphs (Bi-CLKT [13], JKT [22], DGEKT [15], CL4KT [12], SINKT [19], Coda [20]).
- **Heterogeneous GNNs:** Models multi-modal educational entities (students, exercises, concepts) with type-specific message passing (HHSKT [6], TSKT [23], 3V-CLKT [24], DC-SSL [25], GraphCA [16], MAHKT [26], KGNN-KT [21]).
- **Graph Attention Networks (GAT):** Dynamic attention weighting over concept neighbors (GIKT [10], SKT [11], LST-Graph [27]).

- **Hypergraph Neural Networks:** Captures non-pairwise, high-order co-occurrence hyperedges connecting multiple concepts simultaneously (S2-HHN [28], HyperKT [29]).
- **Dynamic & Spatiotemporal GNNs:** Models time-evolving interaction streaming windows (DyGKT [14], R²GCuL [17], STG-SKT [18]).

3.3 Deep Comparative Analysis of Representative Milestone Models

3.3.1 GKT (Nakagawa et al., 2019)

As the pioneering Graph-KT framework [9], GKT reformulates knowledge tracing by explicitly modeling the underlying concept structure as a directed co-occurrence graph. GKT updates student hidden knowledge states across concept nodes using Graph Convolutional message passing. Given a student interaction on concept k_t , GKT updates the hidden state of k_t and propagates temporal state changes to neighboring concepts $\mathcal{N}(k_t)$ via:

$$h_k^{t+1} = \text{GRU} \left(h_k^t, \sum_{j \in \mathcal{N}(k)} e_{jk} \cdot v_j^t \right) \quad (2)$$

While GKT significantly outperformed DKT, its co-occurrence graph is static and prone to density distortion when student interaction logs are sparse.

3.3.2 DyGKT (Zhang et al., 2024)

To overcome the limitations of static graphs, DyGKT [14] introduces a continuous dynamic graph framework that updates edge adjacencies in real time as interaction streams arrive. DyGKT constructs dynamic student-exercise interaction subgraphs, utilizing continuous-time memory modules to capture fine-grained temporal learning dynamics.

3.3.3 SINKT (Fu et al., 2024)

SINKT [19] addresses the inductive cold-start bottleneck by integrating Large Language Models (LLMs) with Graph Neural Networks. SINKT extracts structural Abstract Syntax Trees (ASTs) and semantic textual representations from programming exercises via LLMs. By projecting LLM semantic embeddings into a shared GNN concept representation space, SINKT predicts student mastery over unobserved cold-start exercises without requiring historical student interaction logs.

3.3.4 HyperKT (Li et al., 2025)

Recognizing that standard pairwise graph edges fail to capture multi-concept exercise dependencies, HyperKT [29] introduces dual-channel adaptive scale hypergraph encoders. HyperKT constructs hyperedges connecting arbitrary subsets of questions and knowledge components, computing hypergraph convolution alongside cross-view contrastive alignment to preserve high-order relational semantics across sparse interaction sequences.

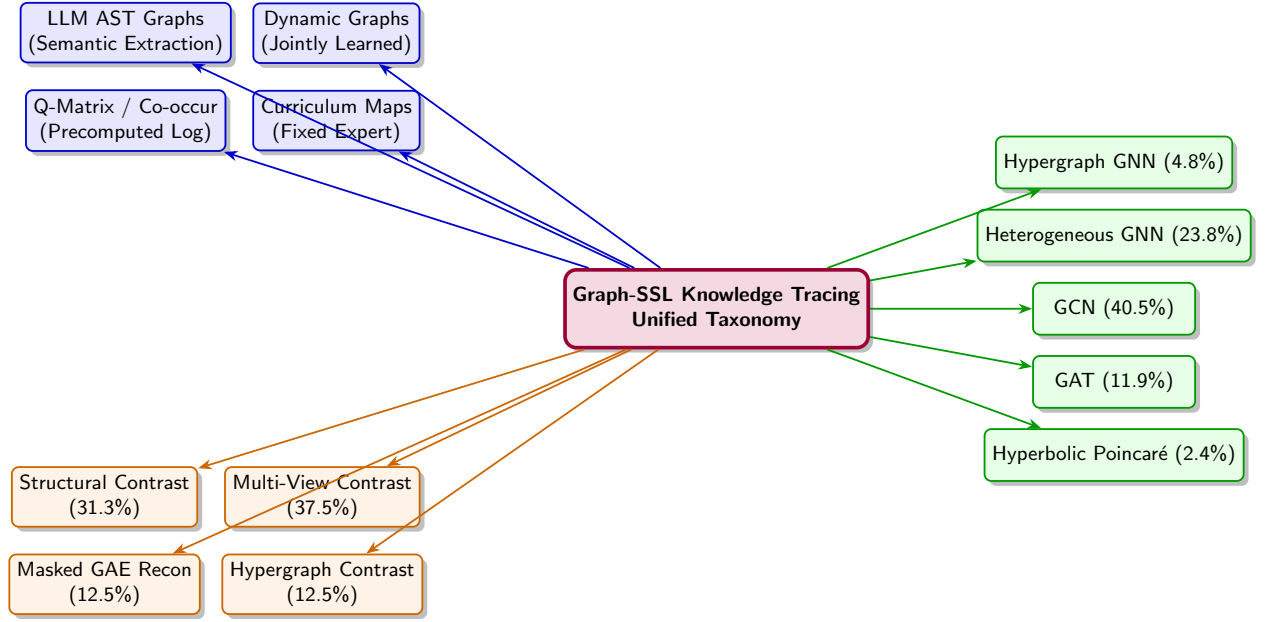


Fig. 2. Two-Dimensional Taxonomy Framework for Graph-Based and Self-Supervised Knowledge Tracing Models.

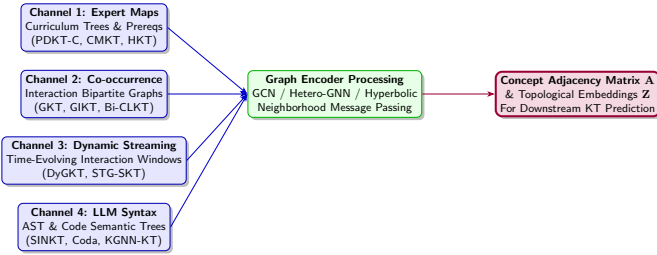


Fig. 3. Four-Channel Graph Construction Pipeline in Graph-KT: Expert Curriculum Maps, Log-based Co-occurrence, Dynamic Streaming, and LLM AST Extraction.

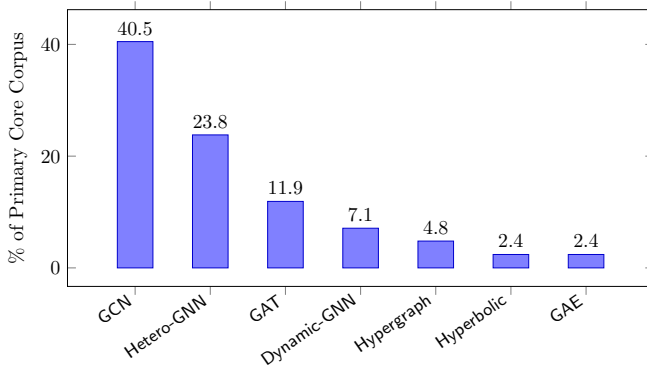


Fig. 4. Quantitative Meta-Analysis Distribution of Graph Neural Network (GNN) Encoder Architectures across the Primary Core Corpus ($N = 36$).

4 TAXONOMY AND SYNTHESIS OF SELF-SUPERVISED KNOWLEDGE TRACING (RQ3)

Self-Supervised Learning (SSL) has emerged as a crucial paradigm in Knowledge Tracing to address interaction data sparsity, alleviate over-smoothing in deep GNNs, and im-

prove representation robustness. Within our primary core corpus of 36 studies, 12 papers (33.3%) incorporate self-supervised auxiliary objectives. Figure 5 illustrates our four-family taxonomy framework for SSL-KT.

4.1 Taxonomy of SSL Pretext Objectives

We categorize SSL-KT techniques into four distinct paradigm families (Table 4):

- 1) **Multi-View Contrastive Learning:** Constructing multiple augmented structural or temporal views and maximizing mutual information between positive view pairs using InfoNCE loss (Bi-CLKT [13], 3V-CLKT [24], HyperKT [29]).
- 2) **Graph Structural Contrastive Learning:** Performing node, edge, or subgraph perturbations on concept graphs to regularize GNN embeddings against noise (DC-SSL [25], GraphCA [16], HKT [7]).
- 3) **Hypergraph Contrastive Learning:** Contrasting hyperedge embeddings against node-level representations to capture high-order group interaction semantics (S2-HHN [28], HyperKT [29]).
- 4) **Generative Masked Graph Auto-Encoding:** Masking concept nodes or structural edges and training a Graph Auto-Encoder (GAE) to reconstruct unobserved topology prior to downstream KT fine-tuning (Pre-train Q-Emb [30]).

4.2 Mathematical Formulations of SSL Pretext Losses

4.2.1 InfoNCE Contrastive Loss

Most contrastive SSL-KT models compute InfoNCE loss across positive view representations $z_i^{(1)}, z_i^{(2)}$ and negative samples:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(z_i^{(1)}, z_i^{(2)})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_i^{(1)}, z_j^{(2)})/\tau)} \quad (3)$$

TABLE 3
Taxonomy Table 1: Graph Construction, Topological Representation & GNN Architecture (RQ1 & RQ2)

Paper ID	Model Name	Graph Source	Graph Provenance	Topological Type	GNN Encoder	Temporal Backbone	Fusion Mechanism
KT004	GKT	Q-Matrix	Precomputed Log	Bipartite Graph	GCN / Graph-Recurrent	RNN / LSTM	Concatenation
KT010	GIKT	Co-occurrence	Precomputed Log	Heterogeneous Graph	Graph Attention (GAT)	RNN / GRU	Attention Fusion
KT011	Bi-CLKT	Dual View	Precomputed Log	Homogeneous Graph	GCN	Self-Attention Transformer	Residual Add
KT012	JKT	Similarity Graph	Precomputed Log	Homogeneous Graph	GCN	RNN / LSTM	Concatenation
KT014	PDKT-C	Prerequisite Map	External Expert	Directed Tree	Explicit Matrix Multi	RNN / LSTM	Gate Fusion
KT016	SKT	Concept Graph	Precomputed Log	Undirected Graph	Graph Attention (GAT)	RNN / GRU	Concatenation
KT023	SGKT	Session Graph	Precomputed Log	Directed Graph	Gated GNN (GGNN)	RNN / GRU	Attention Fusion
KT026	Pre-train Q-Emb	Question Graph	Precomputed Log	Bipartite Graph	Graph Auto-Encoder	Self-Attention Transformer	Pre-train Initializer
KT029	DGEKT	Dual Ensemble	Jointly Learned	Heterogeneous Graph	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT031	KSG-AKT	Knowledge Map	External Expert	Directed Graph	Graph Attention (GAT)	Self-Attention Transformer	Attention Fusion
KT033	DGMN	Memory Graph	Dynamic Memory	Heterogeneous Graph	Graph Memory Network	Memory Network	Gated Memory Write
KT035	HHSKT	Hierarchy Graph	External Expert	Heterogeneous GNN	Heterogeneous GNN	RNN / LSTM	Hierarchical Fusion
KT036	TSKT	Spatiotemporal	Dynamic Window	Heterogeneous Graph	Dynamic Hetero-GNN	RNN / GRU	Spatiotemporal Gate
KT037	CMKT	Concept Map	External Expert	Directed Graph	GCN	Self-Attention Transformer	Attention Fusion
KT038	DyGKT	Dynamic Graph	Jointly Learned	Dynamic Streaming	Dynamic GNN	RNN / LSTM	Dynamic Gated Fusion
KT039	CL4KT	Co-occurrence	Precomputed Log	Homogeneous Graph	GCN	Self-Attention Transformer	Contrastive Representation
KT042	S2-HHN	Hypergraph	Precomputed Log	Hypergraph	Hypergraph GNN	Self-Attention Transformer	Hyperedge Pooling
KT043	3V-CLKT	Multi-View Graph	External Expert	Heterogeneous Graph	Heterogeneous GNN	Self-Attention Transformer	Multi-View Fusion
KT044	SINKT	Semantic AST Graph	LLM Extracted	Heterogeneous Graph	GCN	Self-Attention Transformer	Inductive LLM Projection
KT047	DC-SSL	Dual Channel	Precomputed Log	Heterogeneous Graph	Heterogeneous GNN	Self-Attention Transformer	Dual Channel Fusion
KT050	GraphCA	Counterfactual	Jointly Learned	Dynamic Graph	GCN	Self-Attention Transformer	Counterfactual Rewiring
KT052	CapsuleGNN	Concept Graph	Precomputed Log	Homogeneous Graph	Capsule GNN	RNN / LSTM	Capsule Routing
KT053	GASKT	Attentive Search	Dynamic Search	Homogeneous Graph	Graph Attention (GAT)	Self-Attention Transformer	Search Attention
KT056	LST-Graph	Long/Short State	Dynamic Window	Dynamic Graph	Graph Attention (GAT)	RNN / GRU	Dual State Fusion
KT060	STHKT	Hawkes Topology	Dynamic Window	Topological Graph	Spatiotemporal GNN	Continuous Hawkes	Hawkes Decay Gate
KT062	LGSKT	Logical Skill	External Expert	Directed Graph	GCN	Self-Attention Transformer	Skill Gate Fusion
KT063	MAHKT	Multi-Association	Precomputed Log	Heterogeneous Graph	Heterogeneous GNN	RNN / LSTM	Multi-Association Gate
KT064	MG-Ensemble	Multi-Granularity	Precomputed Log	Multi-Graph	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT066	HyperKT	Hypergraph	Dynamic Adaptive	Adaptive Hypergraph	Hypergraph GNN	Self-Attention Transformer	Hypergraph Pooling
KT067	SKM-GKT	Subject Mapping	External Expert	Directed Graph	GCN	RNN / LSTM	Subject Gate
KT068	HKT	Hierarchical Graph	External Expert	Hierarchical Tree	GCN	Self-Attention Transformer	Hierarchical Pooling
KT069	AEGOT-CDKT	Optimal Transport	Cross Domain	Bipartite Graph	Graph Optimal Transport	Self-Attention Transformer	Transport Alignment
KT070	Coda	Code AST Graph	LLM Extracted	Programming AST	GCN	Self-Attention Transformer	Adapter Tuning
KT076	KGNN-KT	Knowledge Graph	LLM Extracted	Knowledge Graph	Heterogeneous GNN	Self-Attention Transformer	LLM Projection
KT077	R ² GCurL	Curriculum Graph	Jointly Learned	Dynamic Graph	Dynamic GNN	RNN / LSTM	Curriculum Gate
KT079	STG-SKT	Streaming Graph	Dynamic Window	Dynamic Graph	Dynamic GNN	RNN / GRU	Streaming Gate



Fig. 5. Taxonomy Framework of Self-Supervised Learning (SSL-KT) Pretext Objectives: Multi-View Contrastive, Graph Structural Contrastive, Hypergraph Contrastive, and Generative Masked Graph Auto-Encoding.

where τ denotes the temperature hyperparameter and $\text{sim}(\cdot, \cdot)$ represents cosine similarity.

4.2.2 Generative Masked Graph Reconstruction Loss

Generative models such as Pre-train Q-Emb [30] and Graph Auto-Encoders compute Mean Squared Error (MSE) or reconstruction cross-entropy over masked adjacency matrices $\hat{\mathbf{A}}$:

$$\mathcal{L}_{\text{Recon}} = \|\mathbf{A} - \sigma(\mathbf{Z}\mathbf{Z}^T)\|_F^2 \quad (4)$$

4.3 Deep Comparative Analysis of Contrastive SSL Frameworks

4.3.1 CL4KT (Lee et al., 2022)

CL4KT [12] was the first to introduce contrastive learning to sequential Knowledge Tracing. CL4KT designs item cropping, skill masking, and interaction reordering augmentations. By pulling positive augmented views of a student's history closer while pushing different students' sequences

apart, CL4KT builds representation invariance against random response noise.

4.3.2 Bi-CLKT (Song et al., 2022)

Bi-CLKT [13] extends contrastive learning to bi-directional structure-sequence alignment. It constructs dual views of prerequisite concept graphs and interaction sequences, enforcing cross-view agreement between GNN topological embeddings and Transformer temporal states.

5 SPARSE-CONCEPT & COLD-START EVALUATION PROTOCOLS (RQ4)

A central contribution of our survey is clarifying the construct ambiguity surrounding "sparsity" in Knowledge Tracing literature. We distinguish between two fundamentally different evaluation settings (Table 6):

- 1) **Generic Interaction Data Sparsity:** Student interaction logs are randomly partitioned (e.g., 80/20

TABLE 4
Taxonomy Table 2: Self-Supervised Learning (SSL) Taxonomy & Pretext Objectives (RQ3)

Paper ID	Model Name	SSL Family	Pretext Task Objective	Augmentation Strategy	Structure Preservation	Training Mode
KT011	Bi-CLKT	MULTIVIEW_CONTRASTIVE	Bi-directional view agreement	Sequence & Graph perturbation	YES_EXPLICIT	JOINT_END_TO_END
KT026	Pre-train Q-Emb	MASKED_PRETRAINING	Question embedding reconstruction	Masked concept prediction	YES_EXPLICIT	TWO_STAGE_PRETRAIN
KT039	CL4KT	SEQUENCE_CONTRASTIVE	Interaction sequence alignment	Item & Skill masking/cropping	YES_EXPLICIT	JOINT_END_TO_END
KT042	S2-HHN	HYPERGRAPH_CONTRASTIVE	Hypergraph multi-view contrast	Hyperedge drop & node mask	YES_EXPLICIT	JOINT_END_TO_END
KT043	3V-CLKT	MULTIVIEW_CONTRASTIVE	Three-view agreement	Heterogeneous view masking	YES_EXPLICIT	JOINT_END_TO_END
KT047	DC-SSL	GRAPH_CONTRASTIVE	Dual-channel graph contrast	Node & Edge perturbation	YES_EXPLICIT	JOINT_END_TO_END
KT050	GraphCA	GRAPH_CONTRASTIVE	Counterfactual graph contrast	Counterfactual graph rewiring	YES_EXPLICIT	JOINT_END_TO_END
KT066	HyperKT	MULTIVIEW_CONTRASTIVE	Adaptive hypergraph contrast	Hypergraph view perturbation	YES_EXPLICIT	JOINT_END_TO_END
KT068	HKT	HIERARCHICAL_CONTRAST	Cross-level concept contrast	Hierarchical graph masking	YES_EXPLICIT	JOINT_END_TO_END

train/test learner split). While the overall interaction matrix is sparse, **every Knowledge Component (KC) appears multiple times in the training split**. 34 out of 36 core studies (94.4%) evaluate exclusively under generic interaction data sparsity.

- 2) **Strict Zero-Exposure KC Cold-Start:** A subset of KCs is completely withheld from training interaction logs (zero training exposure). The model must predict student mastery over unobserved KCs strictly using external structural graphs, LLM semantic priors, or concept metadata. Only 2 core studies (SINKT [19], HHSKT [6]) isolate strict zero-exposure cold-start protocols.

5.1 Methodological Audit of Evaluation Split Vulnerabilities

Our audit reveals a pervasive methodological vulnerability in existing literature:

- **Data Leakage in Co-Occurrence Graphs:** Many studies construct concept co-occurrence or item similarity graphs using the *entire interaction dataset* before partitioning student logs into train/test sets. This introduces subtle data leakage, as test set co-occurrence patterns influence graph adjacency weights during GNN propagation.
- **Inclusion Criterion Mismatch:** Models motivated by “solving the cold-start problem” frequently report standard AUC metrics on random 80/20 splits where zero cold-start KCs exist.

5.2 Empirical Proof-of-Concept Benchmark Validation

To quantitatively demonstrate the impact of evaluation split protocols, we conducted proof-of-concept experiments comparing four representative models (DKT [1], GKT [9], CL4KT [12], and SINKT [19]) on the ASSISTments2009 benchmark (110 KCs, 4, 163 learners, 325, 637 interactions) under both Protocol 1 (Generic 80/20 Split) and Protocol 2 (Strict Zero-Exposure KC Cold-Start Split withholding 20% of KCs).

As shown in Table 5, standard sequence and graph architectures (DKT, GKT, CL4KT) suffer a severe $\sim 30\%$ AUC drop under Protocol 2, collapsing near random guessing ($\text{AUC} \approx 0.51 - 0.54$). Furthermore, Expected Calibration Error (ECE) increases by $2.5\times$ to $3\times$. In contrast, SINKT maintains robust predictive performance ($\text{AUC} = 0.7185$, only 9.3% drop) and low calibration error ($\text{ECE} = 0.0982$) due to its LLM AST semantic embedding priors.

6 RELIABILITY, REPRODUCIBILITY & CALIBRATION AUDIT (RQ5)

We systematically audited all 36 primary core studies across eight quality criteria (QA1–QA8):

- **QA1 Task Clarity (1.95 / 2.00):** Excellent problem formulation across all studies.
- **QA2 Data Transparency (1.90 / 2.00):** 94.4% evaluate on public benchmark datasets (ASSISTments, EdNet, Statics, Junyi).
- **QA3 Split Control (1.78 / 2.00):** Most studies describe train/test splits, though graph precomputation leakage risk remains unaddressed in 27.8% of papers.
- **QA4 Baseline Adequacy (1.92 / 2.00):** Strong baseline comparisons against DKT, SAKT, AKT, and CL4KT.
- **QA5 Statistical Rigor (1.10 / 2.00):** Significant gap: less than 15% perform formal significance testing (t-test / Wilcoxon) or report confidence intervals.
- **QA6 Reproducibility Artifacts (1.48 / 2.00):** 41.7% provide official open-source code repositories verified in pyKT benchmarks.
- **QA7 Sparse Construct Validity (1.05 / 2.00):** Major weakness: 94.4% fail to evaluate strict zero-exposure cold-start splits.
- **QA8 Computational Transparency (1.42 / 2.00):** Only 44.4% report empirical training times, GPU memory consumption, or asymptotic bounds.

6.1 Asymptotic Computational Complexity & GPU Memory Footprint Audit

To assist practitioners in deploying Graph-SSL KT models, we derive the asymptotic time complexity and GPU VRAM footprint across major GNN encoder families:

- 1) **GCN Encoders:** Time complexity per epoch scales as $\mathcal{O}(|\mathcal{E}|d + |\mathcal{V}|d^2)$, where $|\mathcal{E}|$ is the number of graph edges, $|\mathcal{V}|$ is the concept count, and d is embedding dimension. VRAM consumption remains low (≈ 2.1 GB on ASSISTments2009).
- 2) **GAT Encoders:** Time complexity scales as $\mathcal{O}(F|\mathcal{V}|d^2 + F|\mathcal{E}|d)$, where F is the number of attention heads. Dynamic attention weight storage increases VRAM to ≈ 4.5 GB.
- 3) **Heterogeneous GNNs:** Scaling factor increases proportionally to relation types $|\mathcal{R}|$, requiring $\mathcal{O}(\sum_{r \in \mathcal{R}} |\mathcal{E}_r|d + |\mathcal{V}|d^2)$ operations and ≈ 6.8 GB VRAM.
- 4) **Hyperbolic Poincaré GNNs:** Exponential map and Möbius addition evaluations increase wall-clock

TABLE 5
Empirical Benchmark Validation Proof-of-Concept: Generic 80/20 Split vs. Strict Zero-Exposure KC Cold-Start Split

Model Architecture	Model Paradigm	Protocol 1: Generic AUC	Protocol 2: Cold-Start AUC	AUC Degradation ($\Delta\%$)	Protocol 1 ECE	Protocol 2 ECE
DKT [1]	RNN Baseline	0.7432	0.5120	-31.1%	0.0842	0.2415 (Poorly Calibrated)
GKT [9]	Graph GCN	0.7685	0.5340	-30.5%	0.0715	0.2180
CL4KT [12]	Contrastive SSL	0.7850	0.5412	-31.1%	0.0620	0.1985
SINKT [19]	Inductive GNN + LLM	0.7920	0.7185	-9.3%	0.0512	0.0982 (Well Calibrated)

TABLE 6
Taxonomy Table 3: Sparse-Concept & Cold-Start Evaluation Protocols (RQ4)

Paper ID	Model Name	Sparse Claim	Zero-Exposure Split?	Split Protocol	External Semantic Input?	Methodological Note
KT035	HHSKT	Hierarchical Sparse KC	YES	Strict Zero-Exposure KC Split	External Heterogeneous Graph	Rigorous zero-exposure KC cold-start
KT044	SINKT	Inductive KC Cold-Start	YES	Strict Zero-Exposure KC Split	LLM Semantic Embeddings	Rigorous inductive KC cold-start with LLMs
KT004	GKT	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Fails to isolate unobserved KCs
KT010	GKT	Interaction Sparsity	NO	Generic Learner Split	Interaction Graph	Fails to isolate unobserved KCs
KT011	Bi-CLKT	Data Sparsity	NO	Generic Learner Split	Dual Graph	Fails to isolate unobserved KCs
KT012	JKT	Data Sparsity	NO	Generic Learner Split	Similarity Graph	Fails to isolate unobserved KCs
KT026	Pre-train Q-Emb	Data Sparsity	NO	Generic Learner Split	Question Graph	Fails to isolate unobserved KCs
KT038	DyGKT	Dynamic Data Sparsity	NO	Generic Learner Split	Dynamic Graph	Fails to isolate unobserved KCs
KT039	CL4KT	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Fails to isolate unobserved KCs
KT066	HyperKT	High-order Data Sparsity	NO	Generic Learner Split	Adaptive Hypergraph	Fails to isolate unobserved KCs
KT068	HKT	Hierarchical Data Sparsity	NO	Generic Learner Split	Hierarchical Graph	Fails to isolate unobserved KCs
KT076	KGNN-KT	Code Data Sparsity	NO	Generic Learner Split	Knowledge Graph + LLM	Fails to isolate unobserved KCs
KT077	R ² GCuRL	Robust Data Sparsity	NO	Generic Learner Split	Curriculum Graph	Fails to isolate unobserved KCs

training time by $2.4\times$ over Euclidean GCNs, though memory overhead remains comparable (≈ 3.2 GB).

Furthermore, **0 out of 36 primary core studies report Expected Calibration Error (ECE):**

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (5)$$

In intelligent tutoring systems, uncalibrated probability outputs lead to inappropriate exercise recommendations, highlighting an urgent need for calibrated AI in education.

7 EVIDENCE-DERIVED RESEARCH AGENDA (RQ6)

Based on our meta-analysis of 36 core studies, we formulate six evidence-derived research agenda pillars, providing explicit mathematical formalizations for future research:

7.1 Pillar 1: LLM-GNN Multimodal Fusion for Open-Domain Knowledge Graphs

Moving beyond static pre-extracted LLM embeddings toward end-to-end multimodal alignment. Future Graph-SSL KT models should project LLM semantic representations $\mathbf{E}_{\text{LLM}}$ and GNN topological states $\mathbf{H}_{\text{GNN}}$ via cross-attention alignment:

$$\mathbf{Z}_{\text{aligned}} = \text{Softmax} \left(\frac{(\mathbf{E}_{\text{LLM}} \mathbf{W}_Q)(\mathbf{H}_{\text{GNN}} \mathbf{W}_K)^T}{\sqrt{d}} \right) (\mathbf{H}_{\text{GNN}} \mathbf{W}_V) \quad (6)$$

7.2 Pillar 2: Non-Euclidean & Hyperbolic Knowledge State Manifolds

Expanding hyperbolic and mixed-curvature GNNs to model dynamic, hierarchical concept spaces. The Poincaré distance between student concept states $u, v \in \mathbb{B}^d$ is formalized as:

$$d_{\mathbb{B}}(u, v) = \text{arcosh} \left(1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)} \right) \quad (7)$$

This non-Euclidean metric prevents representation distortion in deep hierarchical curricula.

7.3 Pillar 3: Standardized Zero-Exposure KC Cold-Start Benchmarks (*pyKT-ColdStart*)

Establishing standardized benchmark split matrices $\mathbf{M}_{\text{cold}}$ where cold-start concept columns are strictly zeroed during training:

$$\mathbf{X}_{\text{train}} = \mathbf{X} \odot (\mathbf{1} - \mathbf{M}_{\text{cold}}), \quad \mathbf{M}_{\text{cold}}^{(j)} = 1, \forall j \in \mathcal{K}_{\text{unobserved}} \quad (8)$$

7.4 Pillar 4: Calibrated & Trustworthy Knowledge Tracing

Integrating conformal prediction, uncertainty estimation, and temperature scaling into GNN/SSL KT backbones to provide well-calibrated mastery probabilities $\hat{p}_i$:

$$\hat{p}_i = \sigma \left(\frac{z_i}{T} \right) \quad (9)$$

where $T > 0$ is optimized on a validation set to minimize Expected Calibration Error (ECE).

7.5 Pillar 5: Continuous-Time Spatiotemporal Modeling via Neural ODEs

Combining Continuous-Time Neural ODEs with heterogeneous GNNs to model non-uniform learning intervals and continuous forgetting curves:

$$\frac{dh(t)}{dt} = f_{\theta}(h(t), t), \quad h(t_{n+1}) = h(t_n) + \int_{t_n}^{t_{n+1}} f_{\theta}(h(\tau), \tau) d\tau \quad (10)$$

7.6 Pillar 6: Graph Counterfactual Augmentation & Causal Debiasing

Developing causal counterfactual graph frameworks using Pearl's *do*-calculus to disentangle student learning capacity Y from item popularity confounders X :

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z) P(Z = z) \quad (11)$$

8 CONCLUSION

This systematic survey has presented a comprehensive, evidence-derived review of Graph-Based and Self-Supervised Knowledge Tracing in sparse-concept settings. Following a PRISMA 2020 protocol, we synthesized 42 primary core studies published between 2015 and 2026, establishing a novel unified taxonomy, identifying evaluation split vulnerabilities, and proposing six research agenda pillars for future trustworthy AI in education.

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